DOMAIN 6 STUDY MATERIALS QUESTIONS

Question 1

Question: Which of the following is the most common occupational disease?

- a. Siderosis
- b. Asbestosis
- c. Dermatitis
- d. Glanders

Correct Answer: Select (c)
Explanation: Dermatitis, which is classified into primary irritation and sensitization of skin, is the most frequent occupational disease. It is caused by contact with acids, alkaline materials such as soaps and detergents, fabric softeners, solvents, or other chemicals. The reaction usually looks like a burn.

Other irritants may include:

- Cement
- Hair dyes
- Long-term exposure to wet diapers
- Pesticides or weed killers
- Rubber gloves
- Shampoos

Question 2

Question: Exposure to petroleum products, greases, and solvents account for the most probable cause of which of the following occupational disease?

- a. Dermatitis
- b. Carpal Tunnel Syndrome
- c. Glanders
- d. Siderosis

Correct Answer: Select (a)
Explanation: Many workers simply use hands with no PPE to work with petroleum products, greases, and solvents. As a result, dermatitis is the most common occupational disease caused by these products.

Glanders is an infectious disease that is caused by the bacterium Burkholderia mallei. While people can get the disease, glanders is primarily a disease affecting horses. It also affects donkeys and mules and can be naturally contracted by other mammals such as goats, dogs, and cats.

Siderosis is a lung disease caused by inhaling metal fumes. Mortality studies among iron workers (welders, iron foundry workers, and steel workers) show that an association exists between working with iron and death from lung cancer and other respiratory causes.

Carpal tunnel syndrome occurs when the median nerve, which runs from the forearm into the palm of the hand, becomes pressed or squeezed at the wrist. Symptoms include frequent burning, tingling, or itching numbness in the palm of the hand and other fingers. Occupationally speaking, carpal tunnel syndrome is caused by repeated use of vibrating hand tools.

Question 3

Question: Which of the following statements BEST describes Cumulative Trauma Disorders (CTD)?

- a. Disorders caused as a result of repeated movement of body parts.
- b. Disorders caused as a results of exposure to chemicals over a relative long period of time.
- c. Mental and psychological disorders developed over a period of time as a result of work related stress.
- d. Any disorder, physical or psychological, which is work related and has a cumulative effect over a given period of time.

Correct Answer: Select (a)
Explanation: CTD is a term used to describe physical disorders that are caused as a result of repeated movement of body parts. CTDs result when repeated stress is placed on tendons, muscles, or nerves causing inflammation or damage. Some of the more common CTDs are tendonitis (inflammation of the tendons), tenosynovitis (inflammation of the synovial sheath), and carpal tunnel syndrome (compression of the median nerve). Some of the conditions that may lead to CTDs are repetition, posture, and lack of rest.

Question 4

Question: Assuming that the NIOSH Recommended Weight Limit (RWL) is 16 lb, what is the Lift Index (LI) for lifting a 22 lb weight?

- a. 1.375
- b. 0.727
- c. 1.896
- d. 2.749

Correct Answer: Select (a)
Explanation: $$LI = \frac{22}{16} = 1.375$$

Note that a LI that exceeds 1.0 is too much weight for the worker, and the task must be changed to lower the LI.

Question 5

Question: A pocket dosimeter is a device that is used to measure employee exposure to which of the following?

- a. Radiation
- b. Noise
- c. Vibration
- d. Airborne hazardous chemicals concentration

Correct Answer: Select (a)
Explanation: The correct answer is (a). Pocket dosimeters are used to measure exposure to gamma rays and X-rays. The two types most commonly used in industry are direct read pocket dosimeters (DRPD) and digital electronic dosimeters (DED).

Question 6

Question: A container has dimensions of $20 \times 20 \times 10\text{ feet}$ and contains nitrogen. Assume that 18.9 ft^3 of carbon dioxide (CO_2) is released into this container. The concentration of CO_2 expressed in volume fraction is:

- a. 2.3
- b. 0.006
- c. 0.0047
- d. 0.65

Correct Answer: Select (c)
Explanation: Volume of air in the container initially: $$20 \times 20 \times 10 = 4,000\text{ ft}^3$$ $$\text{total CO}_2 = 18.9\text{ ft}^3$$ $$\text{total volume} = 4,000 + 18.9 = 4,018.9\text{ ft}^3$$ $$\text{Partial volume of CO}2 : \frac{V{\text{CO}2}}{V{\text{total}}} = \frac{18.9}{4,018.9}$$ $$\text{CO}_2\text{ concentration} = 4.7 \times 10^{-3}\text{ volume fraction} = 0.0047$$

Question 7

Question: In the control of heat stress, an increase in air velocity:

- a. Increases heat transfer rates by convection
- b. Decreases heat transfer rates by convection
- c. Heat transfer is only dependent on the temperature of air
- d. Increases heat transfer rate by conduction

Correct Answer: Select (a)
Explanation: Increase in air velocity increases heat transfer by convection as well as evaporative rate of heat transfer. Heat transfer by convention is directly proportional to the temperature difference and area of heat transfer: $$Q = HA(T_1 - T_2)$$ The constant of proportionality H is strongly dependent on the velocity. As velocity increases, so does H resulting in an increase in convective rate of heat transfer.

Question 8

Question: Dilution ventilation is best suited for which of the following situations?

- a. When highly flammable gases are present
- b. When highly toxic air contaminants are present
- c. When air contaminants are not highly flammable or toxic and are not produced at high rates
- d. When contaminants that are toxic or flammable are produced by a fixed source

Correct Answer: Select (c)
Explanation: Dilution ventilation should not be used as a primary control when air contaminants are highly flammable or toxic or when they are produced in high rates. The vapors or gases may collect locally and pose a severe fire or health hazard.

Question 9

Question: Although widely used in the past, the use of air changes per hour as a measure of dilution ventilation has been eliminated from most ventilation standards. The main reason for this is:

- a. It is difficult to calculate
- b. It does not take into account the actual dilution of contaminants
- c. It relies on older technology no longer available
- d. It cannot be measured

Correct Answer: Select (b)
Explanation: Number of air changes per hour does not accurately reflect the actual dilution of air contaminants. For example, a room with a ceiling height of 8 feet will have twice the air changes per hour of a room with a 16 feet ceiling, assuming each has the same air volume flow rate.

Question 10

Question: In an operation 8 pints of acetone is evaporated per hour. Acetone has a TLV of 2 per cent by volume in air and a specific gravity of 0.79. The chemical formula for acetone is $(\text{CH}_3)_2\text{CO}.$ The atomic weight of carbon is 12, hydrogen is 1, and oxygen is 16. The value of general ventilation air flow rate to keep the concentration of acetone at 1.5 percent using a factor of safety of 6 is:

- a. $16,465\text{ ft}^3/\text{min}$
- b. $17,565\text{ ft}^3/\text{min}$
- c. $8,641\text{ ft}^3/\text{min}$
- d. $1,680\text{ ft}^3/\text{min}$

Correct Answer: Select (b)
Explanation: The formula for calculation of general ventilation air flow rate is: $$Q = \frac{(403)(S.G.)(P \times 10^6)(K)}{(M.W.)(L)}$$ Where:

- Q is the required air flow in $\text{ft}^3/\text{min}.$
- $S.G.$ is the specific gravity of solvent
- P is number of pints evaporated every hour
- K is the factor of safety
- $M.W.$ is the molecular weight of solvent
- L is maximum concentration allowed; for example LFL or TLV in parts per million.

Molecular Weight of acetone $= (15) \times 2 + 12 + 16 = 58$ $$Q = \frac{(403)(0.79)(8 \times 10^6)(6)}{(58)(1.5 \times 10^4)}$$ $$Q = 17,565\text{ ft}^3/\text{min.}$$

Question 11

Question: A 55 gallon drum of gasoline has been spilled in a storage area and is evaporating at the rate of 8.0 pints per hour. Assuming that the chemical formula for gasoline is $\text{C}*8\text{H}*{18},$ and that gasoline has a specific gravity of 0.7, what is the required rate of air flow to keep the concentration of gasoline in the air at 1 volume percent? Since gasoline is highly flammable, use a factor of safety of 8 (atomic weight of carbon is 12 and that of hydrogen is 1.)

- a. $16,173\text{ ft}^3/\text{min}$
- b. $1,462\text{ ft}^3/\text{min}$
- c. $875\text{ ft}^3/\text{min}$
- d. $15,837\text{ ft}^3/\text{min}$

Correct Answer: Select (d)
Explanation: $$Q = \frac{(403)(S.G.) \times 10^6 \times P \times K}{(M.W.)(L)}$$ $$M.W. = 8(12) + 18 = 114$$ $$Q = \frac{(403)(0.7) \times 10^6 \times 8 \times 8}{(114)(1) \times 10^4} = 15,837\text{ ft}^3/\text{min}$$

Question 12

Question: A solvent is evaporating at the rate of $0.9\text{ ft}^3/\text{min}$ in a room having dimensions of $20 \times 15 \times 10.$ The ventilation rate in the room is $1,500\text{ ft}^3/\text{min}.$ Assuming STP conditions, the concentration of the solvent vapor in a room after two hours is:

- a. 300 ppm
- b. 600 ppm
- c. 0.06 ppm
- d. 0.5 ppm

Correct Answer: Select (b)
Explanation: This problem can be solved by using the following equation: $$C = \frac{G}{Q}\left(1 - e^{-Qt/V}\right)$$ $$C = \frac{0.9}{1500}\left(1 - e^{-(1500)(120)/3000}\right)$$ $$C = 600\text{ ppm}$$ Where:

- G is the rate of generation of air contaminants in ft^3/min
- Q is the ventilation rate in ft^3/min
- C is the concentration of contaminants in after t minutes
- V is the volume of room in $\text{ft}^3.$ Note: We have multiplied the value of C calculated from the above equation by 10^6 to convert it to parts per million (ppm).

Question 13

Question: The concentration of acetone in a laboratory is 3,000 ppm. If we start the general ventilation in the lab at the rate of $3,000\text{ ft}^3/\text{min}$ and the lab has a volume of $10,000\text{ ft}^3,$ the concentration of acetone after 20 minutes is:

- a. 11.0 ppm
- b. 7.4 ppm
- c. 16.2 ppm
- d. 24.7 ppm

Correct Answer: Select (b)
Explanation: The following equation can be used to solve this problem: $$\text{Ln}\frac{C_2}{C_1} = -\frac{Q}{V}(t_2 - t_1)$$ where:

- C_2 is concentration after t_2 minutes form start of ventilation
- C_1 is concentration at the start of ventilation
- Q is rate of ventilation in ft^3/min
- V is the volume of room in $\text{ft}^3.$ $$\text{Ln}\frac{C_2}{3000} = -\frac{3000}{10000}(20)$$ $$\text{Ln}\frac{C_2}{3000} = -6$$ $$\frac{C_2}{3000} = e^{-6}$$ $$C_2 = 7.4\text{ ppm.}$$

Question 14

Question: How long would it take to reduce the concentration of contaminants from 1,000 ppm to 50 ppm in a room which has a volume of $15,000\text{ ft}^3$ and a general ventilation rate of $2,500\text{ ft}^3/\text{min}?$

- a. 24 minutes
- b. 12 minutes

- c. 18 minutes
- d. 35 minutes

Correct Answer: Select (c)
Explanation: $$\text{Ln}\frac{C_2}{C_1} = -\frac{Q}{V}(t_2 - t_1)$$ $$\text{Ln}\frac{50}{1000} = -\frac{2500}{15000}(t_2 - t_1)$$ $$(t_2 - t_1) = 18\text{ minutes}$$

Question 15

Question: The rate of dilution ventilation to reduce the concentration of contaminants from 5,000 ppm to 50 ppm in 10 minutes in a room having a volume of $7,000\text{ ft}^3$ is:

- a. $6,745\text{ ft}^3/\text{min}$
- b. $3,223\text{ ft}^3/\text{min}$
- c. $5,462\text{ ft}^3/\text{min}$
- d. $7,293\text{ ft}^3/\text{min}$

Correct Answer: Select (b)
Explanation: $$\text{Ln}\frac{C_2}{C_1} = -\frac{Q}{V}(t_2 - t_1)$$ Solve the above equation for Q: $$Q = \frac{-V \text{Ln}\frac{C_2}{C_1}}{t_2 - t_1}$$ $$Q = \frac{-7000 \text{Ln}\frac{50}{5000}}{10}$$ $$Q = 3,223\text{ ft}^3/\text{min}$$

Question 16

Question: When designing or operating a local exhaust system, one important factor is:

- a. That the system fails if any fugitive emissions occur.
- b. Design to remove all emitted contaminants.
- c. The exhaust air volume flow rate must be proportional to the contaminant generation rate.
- d. Local exhaust systems do not provide dilution ventilation.

Correct Answer: Select (b)
Explanation: As opposed to dilution ventilation, local exhaust ventilation tries to capture, contain, and remove all emitted contaminants before dilution with workroom air. All local exhaust systems experience fugitive emissions. The air volume flow rate is based on physical conditions at the hood, not on contaminant generation rate.

Question 17

Question: Air is flowing in a ventilation duct with a velocity of 1,000 feet per minute. The hood static pressure is -2.0 inches of water. The total pressure is:

- a. -2.60 inches of water
- b. -1.94 inches of water
- c. +2.60 inches of water
- d. -0.80 inches of water

Correct Answer: Select (b)
Explanation: In fluid flow, the total fluid pressure has two components, static pressure and velocity pressure. The static pressure is the pressure exerted to the walls of the piping system if the fluid was stationary. Static pressure is

exerted in all directions. Velocity pressure is the pressure due to kinetic motion of the fluid and is exerted in the direction of fluid flow.

The relationship between total pressure (TP), static pressure (SP), and velocity pressure (VP) is: $$TP = SP + VP$$ The relationship between air velocity and velocity pressure is: $$V = 4005 \sqrt{VP}$$ where V is air velocity in feet per minute, and VP is velocity pressure in inches of water.

For the problem at hand: $$VP = \left(\frac{1000}{4005}\right)^2 = 0.06\text{ inches of water}$$ $$TP = -2.0 + 0.06 = -1.94$$ Note: Static pressure can have a positive or negative value. Velocity pressure, on the other hand, always has a positive value.

Question 18

Question: In the diagram shown, the velocity pressure in the 6 inch duct is (assume STP): (Q1 = 200 cfm in 6 in pipe branch, Q2 = 300 cfm in main duct branch)

- a. -0.302 inches of water
- b. +0.762 inches of water
- c. -0.108 inches of water
- d. +0.404 inches of water

Correct Answer: Select (d)
Explanation: The total volumetric flow rate in the main duct is: $$Q = Q_1 + Q_2 = 300 + 200 = 500\text{ ft}^3/\text{min}$$ $$Q = VA$$ Where Q is volumetric flow rate, V is velocity and A is the area of flow. $$500 = V \times \frac{3.14 (6)^2}{4 \times 144}$$ $$V = 2,548\text{ ft}/\text{min.}$$ To obtain velocity pressure we use $V = 4005 \sqrt{VP}$ $$2548 = 4005 \sqrt{VP} \implies VP = 0.404\text{ inches of water.}$$

Question 19

Question: The root mean square sound pressure for a sound wave is 0.4 pascals. The sound pressure level in decibels relative to 20 micropascals is:

- a. 80
- b. 90
- c. 75
- d. 86

Correct Answer: Select (d)
Explanation: The following relationship can be used to calculate the decibel value of a sound wave from root mean square values. $$\text{L(decibels)} = 20 \log \frac{P}{20}$$ where P is in micropascals and 20 micropascals is our reference point. $$L = 20 \log \frac{0.4 \times 10^6}{20} = 86\text{ dB}$$

Question 20

Question: What is the noise dose for a worker who is exposed to a 95 dBA source for a period of 5 hours?

- a. 165%
- b. 125%
- c. 93%

- d. 315%

Correct Answer: Select (b)
Explanation: The noise dose can be calculated from: $$D = 100 \sum_{i=1}^n \frac{C_i}{t_i}$$ where C_i is the exposure time and t_i the allowable time. The allowable time can be calculated from: $$t = \frac{8}{2^{(L-90)/5}}$$ where L is the exposure in decibels. For the problem at hand: $$t = \frac{8}{2^{(95-90)/5}} = 4\text{ hours}$$ $$D = 100 \left(\frac{5}{4}\right) = 125%\text{ of 8 hr. TWA.}$$

Question 21

Question: A worker is exposed to noise level of 110 dBA for one hour, 95 dBA for 3 hours, and 90 dBA for 3 hours. The noise dose for this worker is:

- a. 500%
- b. 275%
- c. 425%
- d. 312%

Correct Answer: Select (d)
Explanation: First, we calculate allowable times for noise levels of 110, 95, and 90 dB: $$t = \frac{8}{2^{(L-90)/5}}$$ $$t_{110} = \frac{8}{2^{(110-90)/5}} = 0.5\text{ hr}$$ $$t_{95} = \frac{8}{2^{(95-90)/5}} = 4\text{ hrs}$$ $$t_{90} = \frac{8}{2^{(90-90)/5}} = 8\text{ hrs}$$ $$\text{noise dose} = 100 \sum_{i=1}^n \frac{C_i}{t_i}$$ $$D = \left(\frac{1}{0.5} + \frac{3}{4} + \frac{3}{8}\right) 100 = 312%\text{ of 8 hour TWA.}$$

Question 22

Question: Two machines each generate a noise level of 80 dBA. When both machines work simultaneously, the noise level is approximately:

- a. 85 dBA
- b. 83 dBA
- c. 86 dBA
- d. 160 dBA

Correct Answer: Select (b)
Explanation: When in a free field and the difference between two sound levels is zero, we add 3 dB to obtain the combined effect.

Question 23

Question: In a machine shop, a worker is exposed to the following levels of noise: 91 dBA (3 hours), 105 dBA (1 hour), 98 dBA (2 hours). The noise dose expressed in percentage of the OSHA permissible exposure is:

- a. 423 %
- b. 87 %
- c. 219 %
- d. 215 %

Correct Answer: Select (c)
Explanation: First, calculate allowable times: $$t_{91} = \frac{8}{2^{(91-90)/5}} = 7\text{ hours}$$ $$t_{98} = \frac{8}{2^{8/5}} = 2.6\text{ hours}$$ $$t_{105} = \frac{8}{2^{15/5}} = 1\text{ hour}$$ $$\text{noise dose} = 100 \left(\frac{3}{7} + \frac{1}{1} + \frac{2}{2.6}\right) = 219%\text{ of 8 hr TWA}$$

Question 24

Question: A square ventilation duct 2 feet on each side connects two rooms through a 20 feet length of the duct. A noise reduction of 0.2 dB/ft of duct is required. What should the sound absorption coefficient of the lining material be?

- a. 0.9176
- b. 0.612
- c. 0.526
- d. 0.186

Correct Answer: Select (d)
Explanation: We use the following equation: $$NR = \frac{12.6 P \alpha^{1.4}}{A}$$ Where NR is the noise reduction in decibels per foot, P is the perimeter of the duct in inches, and A is the cross section area of the duct in square inches. For the problem at hand: $$0.2 = \frac{12.6 \times 8 \times 12 \times \alpha^{1.4}}{24 \times 24}$$ $$\alpha = 0.186$$

Question 25

Question: What is the noise reduction in decibels per foot of a 2 ft. diameter circular duct lined with a material having a sound absorption coefficient of 0.75?

- a. 1.4
- b. 2.3
- c. 3.8
- d. 2.9

Correct Answer: Select (a)
Explanation: $$NR = \frac{12.6 P \alpha^{1.4}}{A}$$ $$NR = \frac{12.6 \times 3.14 \times 24 \times 12 \times 0.75^{1.4}}{3.14(1)^2 \times 144}$$ $$NR = 1.4\text{ decibel/foot}$$

Question 26

Question: A sound absorbing material with an absorption coefficient of 0.8 is used as lining in a circular duct and provides a noise reduction of 3.0 dB per foot. The diameter of the circular duct is:

- a. 24 inches
- b. 6 inches
- c. 15 inches
- d. 12 inches

Correct Answer: Select (d)
Explanation: $$NR = \frac{12.6 P \alpha^{1.4}}{A}$$ Perimeter $P = 3.14 D$ Therefore: $$3 = \frac{12.6(3.14

D)(0.8)^{1.4}}{3.14 D^2 / 4}$$ $$9.42 D^2 = 115.79 D \implies D = 12.29\text{ inches}$$ Since the diameter of the duct cannot be zero, we accept the other solution for D, which is 12.29 inches (rounded to 12 inches).

Question 27

Question: A sound wave has a sound pressure level of 95 dB with respect to 20 micropascals. The root mean square sound pressure of this wave expressed in units of pascal is:

- a. 9,500
- b. 5.2
- c. 1.12
- d. 0.8

Correct Answer: Select (c)
Explanation: $$\text{SPL in decibels} = 20 \log \frac{\text{rms sound pressure}}{20\text{ microPascal}}$$ $$95 = 20 \log \frac{P}{20}$$ $$\log \frac{P}{20} = 4.75 \implies P = 20(56,234) = 1,124,682\text{ micropascals} = 1.12\text{ Pascals}$$

Question 28

Question: The cochlea is the main element of the:

- a. Inner ear
- b. Middle ear
- c. Outer ear
- d. Tympanic membrane

Correct Answer: Select (a)
Explanation: The cochlea is the main element of the inner ear. It is filled with a fluid and shaped like a coil. Sound waves travel through the cochlea before reaching the hair-like nerve cells for transmission to the brain.

Question 29

Question: Noise Induced Hearing Loss (NIHL) results from over-exposure to high levels of noise. NIHL depends on sound intensity and the exposure time. Which of the following statements is generally true regarding NIHL?

- a. High frequency noise is more damaging
- b. Low frequency noise is more damaging
- c. NIHL depends only on the sound intensity and time of exposure
- d. Only (b) and (c) above

Correct Answer: Select (a)
Explanation: Although there are some differences among individuals, exposure to high frequency noise is more damaging to the ears than a low frequency noise having the same sound intensity and for the same exposure time. NIHL is caused by damage to the microscopic hair cells of the cochlea. A sound level of 85 dB or higher can cause permanent damage to hearing, especially if it has a relatively high frequency.

Question 30

Question: Temporary Threshold Shift (TTS) in hearing is caused as a result of damage to which of the following?

- a. Cochlea
- b. Incus
- c. Organ of corti
- d. Tympanic membrane

Correct Answer: Select (c)
Explanation: TTS is caused as a result of over-exposure to loud noise which damages the organ of Corti and its hair cells.

The organ of Corti is located in the inner ear and acts like a microphone for the body. It is situated in one of the three parts of the cochlea and contains four rows of hair cells on its surface. There are about 16,000-20,000 hair cells in the basilar membrane following the spiral of the cochlea. Small movements of the layers of membrane trigger the hair cells, which in turn send a very small voltage to the associated nerve fiber, called an axon. These impulses are transferred to the brain for processing.

Question 31

Question: In a machine shop, a worker is exposed to 90 dBA (1 hr), 110 dBA (5 min), 95 dBA (2 hrs). Which of the following statements is true?

- a. The shop is not in compliance with OSHA noise exposure requirements.
- b. We need additional data to calculate 8-hour TWA sound level for OSHA requirements.
- c. The shop is in compliance with OSHA requirements.
- d. The shop is not in compliance with OSHA requirements. However, use of hearing protectors would bring the shop in compliance.

Correct Answer: Select (c)
Explanation: First calculate OSHA allowable times from: $$t = \frac{8}{2^{(L-90)/5}}$$ $$t_{90} = 8\text{ hrs}, \quad t_{95} = 4\text{ hrs}, \quad t_{110} = 0.5\text{ hr}$$ Calculate noise dose from: $$D = 100 \left(\frac{1}{8} + \frac{2}{4} + \frac{5/60}{0.5}\right) = 79%\text{ of 8 hr TWA.}$$ Since the noise dose percentage does not exceed 100%, this shop is in compliance with OSHA requirements.

Question 32

Question: Raynaud's Syndrome is an occupational disease caused as a result of:

- a. Over-exposure to excessive noise
- b. Over-exposure of hands to vibration
- c. Repetitive movement of back muscles
- d. Over-exposure to ultraviolet radiation.

Correct Answer: Select (b)
Explanation: Raynaud's syndrome, also known as "Dead Fingers" or "White Fingers", results from over-exposure of hands to vibrating tools. The circulation in the fingers of the hand guiding the tool becomes impaired and when these fingers are exposed to cold temperatures, they become white and devoid of sensation.

Question 33

Question: Which of the following statements is true regarding radioisotopes of a given material which are capable of radioactive decay?

- a. Same atomic number but different atomic mass
- b. Same atomic number and same atomic mass
- c. Different atomic number, different atomic mass
- d. Different atomic number, same atomic mass

Correct Answer: Select (a)
Explanation: Radioisotopes are unstable isotopes of a given element that can undergo radioactive decay and emit ionizing radiation. Isotopes of a given element have the same atomic number (number of electrons and protons), but different atomic mass (number neutrons plus number of protons).

Question 34

Question: Which of the following statements is true regarding Radon $^{222}\text{Rn}^{86}$?

- a. It has 86 protons and 222 neutrons
- b. It has 86 electrons and 222 neutrons
- c. Has 86 protons and 222 electrons
- d. Has 86 protons and 136 neutrons

Correct Answer: Select (d)
Explanation: 222 is the atomic mass which is the number of protons (which is equal to number of electrons) plus the number of neutrons. 86 is the number of protons in the nucleus which is equal to number of electrons.
$$\text{number of neutrons} = \text{atomic mass} - \text{number of protons} = 222 - 86 = 136$$

Question 35

Question: The half-life of Radon $^{222}\text{Rn}^{86}$ is 3.8 days. If we start with one gram of radon, how many days would it take to reduce the weight to 1/16 of a gram?

- a. 7.6
- b. 11.4
- c. 15.2
- d. 19.0

Correct Answer: Select (c)
Explanation: | Times (day) | Weight (gram) | | :--- | :--- | | 0.0 | 1 | | 3.8 | 1/2 | | 7.6 | 1/4 | | 11.4 | 1/8 | | 15.2 | 1/16 |

Question 36

Question: The half-life of Radon $^{222}\text{Rn}^{86}$ is 3.8 days. The decay constant for radon expressed in units of sec^{-1} is:

- a. 2.1×10^{-6}
- b. 6.3×10^{-5}

- c. 8.4×10^{-7}
- d. 7.3×10^{-4}

Correct Answer: Select (a)
Explanation: We use the equation $I = I_0 e^{-\mu x}$ Half-life is the length of time which is needed to reduce the activity to one half the original value: $$\frac{1}{2} I_0 = I_0 e^{-\mu(3.8)} \implies e^{-\mu(3.8)} = 0.5$$ $$\text{Ln}(0.5) = -\mu(3.8) \implies -0.693 = -\mu(3.8) \implies \mu = 0.182\text{ days}^{-1}$$ $$\mu = \frac{0.182}{\text{day}} \times \frac{1\text{ day}}{24\text{ hours}} \times \frac{1\text{ hour}}{3600\text{ sec}} = 2.1 \times 10^{-6}\text{ sec}^{-1}.$$

Question 37

Question: The decay constant for a radioactive material is $2.0 \times 10^{-10}\text{ sec}^{-1}.$ How long would it take to reduce the activity of this material to 15 percent of its original value?

- a. 600 years
- b. 300 years
- c. 100 years
- d. 1,000 years

Correct Answer: Select (b)
Explanation: $$I = I_0 e^{-\mu x}$$ $$0.15 I_0 = I_0 e^{-2 \times 10^{-10} x}$$ $$\text{Ln}(0.15) = -2 \times 10^{-10} x \implies x = 9,485,599,924\text{ seconds} \approx 300\text{ years}$$

Question 38

Question: The decay constant for a radioactive material is $1.4 \times 10^{-8}\text{ sec}^{-1}.$ The half-life of this material is:

- a. 3.52 years
- b. 7.89 years
- c. 12.63 years
- d. 1.57 years

Correct Answer: Select (d)
Explanation: $$I = I_0 e^{-\mu x}$$ $$\frac{1}{2} I_0 = I_0 e^{-1.4 \times 10^{-8} x_{1/2}}$$ $$1.4 \times 10^{-8} x_{1/2} = 0.693$$ $$x_{1/2} = \frac{0.693}{1.4 \times 10^{-8}} = 49.5 \times 10^6\text{ seconds} = 1.57\text{ years}$$

Question 39

Question: The amount of radiation emitted by a source at 2 feet away from the source is 0.06 rem. What level of radiation we expect to measure at 6 feet away from the source?

- a. 0.02 rem
- b. $6.66 \times 10^{-3}\text{ rem}$
- c. 0.12 rem
- d. 0.18 rem

Correct Answer: Select (b)
Explanation: Radiation intensity follows the inverse square law: $$I_1 d_1^2 = I_2 d_2^2$$ $$(0.06)(2)^2 = I_2 (6)^2 \implies I_2 = 6.66 \times 10^{-3}\text{ rem}$$

Question 40

Question: The level of radiation at 2 feet from an X-ray machine is A rem. At what distance from the source would the radiation level drop to 0.1A?

- a. 40 ft
- b. 12 ft
- c. 6.32 ft
- d. 20 ft

Correct Answer: Select (c)
Explanation: $$I_1 d_1^2 = I_2 d_2^2$$ $$A(4) = 0.1A d_2^2 \implies d_2^2 = 40 \implies d_2 = 6.32\text{ ft}$$

Question 41

Question: A Geiger-Mueller radiation measuring instrument is suitable for measuring which of the following types of ionizing radiation?

- a. Gamma rays
- b. Alpha rays
- c. Beta rays
- d. All of the above

Correct Answer: Select (d)
Explanation: Geiger-Mueller is suitable for measuring gamma, beta, and alpha rays. It is a tube which is filled by a gas capable of undergoing ionization. Interaction of ionizing radiation with the gas produces ions which in turn produce an electric current which is measured.

Question 42

Question: Luminescent radiation detectors are suitable for measuring which type of ionizing radiation?

- a. Beta rays
- b. Gamma rays
- c. Laser radiation
- d. Alpha rays

Correct Answer: Select (b)
Explanation: Luminescent detectors are effective measuring instruments for flow of neutrons. The collision of neutrons with the solid material in the detector generates excess energy which is emitted in the form of light which is proportional to the amount of energy received by the material.

Question 43

Question: Which of the following (s) is (are) examples of somatic effects of ionizing radiation?

- a. Cancer
- b. Reddening of skin
- c. Cataracts
- d. All of the above.

Correct Answer: Select (d)
Explanation: All are examples of somatic effects of ionizing radiation.

Question 44

Question: Which of the following is most likely to affect the radioactive properties of a radioactive material?

- a. Nuclear reaction
- b. Boiling
- c. Melting
- d. Sublimation

Correct Answer: Select (a)
Explanation: Physical processes such as those mentioned do not usually affect the radioactive properties of a material. Chemical transformations, on the other hand, can drastically affect such properties. For example, one area for handling nuclear waste which is under research is to use nuclear reactions to convert radioisotopes of long half-life to radioisotopes having very short half-life.

Question 45

Question: Erythema is caused by:

- a. Over-exposure to beta rays
- b. Over-exposure to sensitizers
- c. Over-exposure to materials having high flash point
- d. Over-exposure to sunlight

Correct Answer: Select (d)
Explanation: Erythema, which is reddening of skin, sometimes accompanied by blisters is caused as a result of over-exposure to ultraviolet radiation in the sunlight.

Question 46

Question: Scotoma which is loss of vision in a portion of the visual field is most likely to be caused by which of the following types of non-ionizing radiation?

- a. RF radiation
- b. Microwave
- c. Infrared
- d. Ultraviolet

Correct Answer: Select (c)
Explanation: Scotoma is caused as a result of over-exposure to infrared radiation. Extensive over-exposure can result in cataracts.

Question 47

Question: LD50 is:

- a. The chemical dosage which causes death in 50 per cent of a selected population of laboratory animals.
- b. Used only for toxicity representation when the route of entry of the chemical is by ingestion or skin absorption.
- c. Expressed in units of chemical weight per body weight.
- d. All of the above.

Correct Answer: Select (d)
Explanation: Toxicity data is obtained by conducting experiments on selected laboratory test animals. Lethal Dose (LD) is defined as the amount of chemical which can produce death in tested animals. Lethal Dose 50 indicates the amount of chemical which can produce death in 50 percent of tested animals in a given period of time. When the route of entry of the chemical is by ingestion or skin absorption, lethal doses are usually expressed as LD50, and when the route of entry of the chemical into an animal's body is by inhalation, lethal doses are expressed as LC50.

Question 48

Question: When the route of entry of a chemical into the body is by the inhalation of a gas or vapor, the toxicity information is expressed as:

- a. LD50 and PEL
- b. LC50
- c. Toxicity by inhalation (TI)
- d. Route of entry has no effect on presentation of toxicity data

Correct Answer: Select (b)
Explanation: See description of lethal doses in the text.

Question 49

Question: Simple asphyxiants are those chemicals which:

- a. Can interfere with normal hemoglobin function and deprive the body of oxygen.
- b. Cause suffocation by reducing oxygen concentration.
- c. Are inert gases but once in the body are more reactive with blood than oxygen.
- d. Only (a) and (c) above.

Correct Answer: Select (b)
Explanation: Simple asphyxiants can cause suffocation as a result of reducing the available oxygen concentration. For example, if an inert non-toxic gas such as nitrogen is continuously pumped into a room that initially has normal oxygen concentration, eventually the oxygen concentration would diminish to a point that can no longer support life. Argon or methane also are good examples of simple asphyxiants.

Question 50

Question: Chemicals that attack the liver upon entry into the body are called:

- a. Kiverosis
- b. Nephrotoxins
- c. Neurotoxins
- d. Hepatotoxins

Correct Answer: Select (d)
Explanation: Hepatotoxins attack the liver when they enter the blood stream. Examples are carbon tetrachloride, ethyl alcohol, and vinyl chloride.

Question 51

Question: If the air contains 500 ppm of acetone (TLV, 750 ppm), and 50 ppm of MEK (TLV, 200 ppm):

- a. A Self Contained Breathing Apparatus (SCBA) must be used.
- b. The mixture TLV is above the recommended values.
- c. The concentration of the mixture is below recommended values of TLV.
- d. Only the TLV of the most hazardous component must be taken into account.

Correct Answer: Select (c)
Explanation: $$\frac{C_1}{T_1} + \frac{C_2}{T_2} < 1$$ $$\frac{500}{750} + \frac{50}{200} = 0.667 + 0.25 = 0.917 < 1$$ Since the sum of ratios is less than 1, the mixture concentration is below recommended values of TLV.

Question 52

Question: Air contains 700 ppm of methane $(\text{CH}_4).$ How many grams of methane are in a room that has a volume of 25 cubic meters? The atomic weight of carbon is 12 and that of hydrogen is 1.

- a. 16
- b. 21
- c. 11.5
- d. 5

Correct Answer: Select (c)
Explanation: Molecular weight of $\text{CH}_4 = 12 + 4 \times 1 = 16$ $$\text{ppm} = \frac{(\text{mg/m}^3)(24.4)}{\text{M.W.}}$$ $$700 = \frac{(\text{mg/m}^3)(24.4)}{16} \implies \text{mg/m}^3 = 459\text{ mg/m}^3$$ $$\text{total} = 459 \times 25 = 11,475\text{ mg} = 11.475\text{ g} \approx 11.5\text{ grams}$$

Question 53

Question: If the wavelength of an electromagnetic radiation is 10 mm, its period is:

- a. $1 \times 10^{-9}\text{ sec}$

- b. $3 \times 10^{-6}\text{ sec}$
- c. $4.2 \times 10^{-8}\text{ sec}$
- d. $3.3 \times 10^{-11}\text{ sec}$

Correct Answer: Select (d)
Explanation: $$c = \frac{\lambda}{T}$$ Where:

- λ = wavelength $= 10\text{ mm} = 10 \times 10^{-3}\text{ m}$
- T = period
- c = speed of light $= 300,000,000\text{ m/sec} = 3 \times 10^8\text{ m/sec}$ $$T = \frac{\lambda}{c} = \frac{10 \times 10^{-3}\text{ m}}{3 \times 10^8\text{ m/sec}} = 3.3 \times 10^{-11}\text{ sec}$$

Question 54

Question: The exposure rate from a radiation source is 4 mrem/hr. In order to reduce exposure, 4 inch of shielding with a buildup factor of 3.00 and an attenuation coefficient of 0.6 cm^{-1} is added to the source. The intensity of attenuated radiation is:

- a. 3.126 mrem/hr
- b. 6.725 mrem/hr
- c. 0.027 mrem/hr
- d. 8.623 mrem/hr

Correct Answer: Select (c)
Explanation: We use the following relationship: $$I = \beta I_0 e^{-\mu x}$$ where:

- I = intensity of attenuated radiation
- (\beta) = buildup factor, dimensionless (3.00)
- I_0 = intensity of exposure (4 mrem/hr)
- (\mu) = attenuation coefficient (0.6 cm^{-1})
- x = thickness of shield, cm ($4\text{ inches} \times 2.54\text{ cm/in} = 10.16\text{ cm}$) $$I = (3.00)(4) e^{-(0.6 \times 4 \times 2.54)}$$ $$I = 0.027\text{ mrem/hr}$$

Question 55

Question: A source is emitting electromagnetic radiation with a wavelength of 5 cm and an average power of 3,000 W. If the antenna has an area equal to $1\text{ m}^2,$ the power density at 100 meters from the antenna is:

- a. 300 W/m^2
- b. 120 W/m^2
- c. 30 W/m^2
- d. 600 W/m^2

Correct Answer: Select (b)
Explanation: The relationship we use to solve this problem is: $$W = \frac{A P}{\lambda^2 r^2}$$ where:

- W = power density, units of power per distance squared e.g. W/m^2
- A = antenna area (1 m^2)
- P = power (3,000 W)
- r = distance from antenna (100 m)
- (\lambda) = wavelength ($5\text{ cm} = 0.05\text{ m}$) $$W = \frac{1 \times 3000}{(0.05)^2(100)^2} = 120\text{ W/m}^2$$

Question 56

Question: A gamma source with energy of 3.5 MeV has a Curie strength of 400 millicurie. The expected radiation level at a distance of 1 ft. from this source is equal to:

- a. 6.2 R/hr
- b. 2.3 R/hr
- c. 12.5 R/hr
- d. 8.4 R/hr

Correct Answer: Select (d)
Explanation: We use the following relationship to solve this problem: $$S = 6 C E$$ where:

- S = expected radiation level at 1 ft. in roentgens per hour (R/hr.)
- C = strength of radiation source expressed in units of Curie (Ci) ($400\text{ mCi} = 0.4\text{ Ci}$)
- E = energy of gamma radiation expressed in Mev (3.5 Mev) $$S = 6(0.4)(3.5) = 8.4\text{ R/hr.}$$

Question 57

Question: The sound pressure level at a distance of 4 ft. from a sound source is 96 decibels. The sound pressure level at 12 ft. away from this sound source is:

- a. 90 decibels
- b. 82 decibels
- c. 86 decibels
- d. 94 decibels

Correct Answer: Select (c)
Explanation: The relationship between sound level and distance is expressed by the following relationship: $$dB_1 = dB_0 + 20 \log \frac{d_0}{d_1}$$ Where:

- dB_1 = sound level at distance d_1
- dB_0 = sound level at distance d_0 For the problem at hand, we have: $$dB_1 = 96 + 20 \log \frac{4}{12} = 96 - 9.54 = 86.45 \approx 86\text{ decibels}$$

Question 58

Question: If the distance from a sound source is doubled, the reduction in sound pressure level is:

- a. 4 decibels
- b. 8 decibels
- c. 12 decibels
- d. 6 decibels

Correct Answer: Select (d)
Explanation: We start with: $$dB_1 = dB_0 + 20 \log \frac{d_0}{d_1}$$ In this case $\frac{d_0}{d_1} = 0.5$ distance doubled. We have to calculate $dB_1 - dB_0$: $$dB_1 - dB_0 = 20 \log 0.5 = -6\text{ dB}$$ This means that if we double our distance from a sound source, the sound pressure level decreases by 6 decibels.

Question 59

Question: In industrial noise control, every attempt must be made to reduce noise level by means of engineering controls. One way to accomplish this is treatment of ceilings with sound absorbing materials such as fiberglass. In a machine shop with a ceiling area of $6,000\text{ ft}^2$ and sound absorption coefficient of 0.04, the ceiling was treated with a sound absorbing material which has a coefficient of absorption of 0.8. The amount of noise reduction in this machine shop is:

- a. -13 decibels
- b. 48 decibels
- c. 24 decibels
- d. 8 decibels

Correct Answer: Select (a)
Explanation: The relationship for calculation of noise reduction is: $$dB = 10 \log\left(\frac{A_1}{A_2}\right)$$ where:

- dB = amount of noise reduction in decibels
- A_1 = amount of absorption before treatment expressed in units of sabins
- A_2 = amount of absorption after treatment in sabins Values of A_1 and A_2 can be obtained by multiplying the ceiling area by the coefficients of absorption. For the problem at hand: $$A_1 = (6,000\text{ ft}^2) \times (0.04) = 240\text{ sabins}$$ $$A_2 = (6,000\text{ ft}^2) \times (0.8) = 4,800\text{ sabins}$$ $$dB = 10 \log \frac{240}{4800} = -13\text{ decibels}$$

Question 60

Question: The time weighted average noise exposure for a worker is 95 decibels in a shift. The noise dose is:

- a. 120%
- b. 199%
- c. 50%
- d. 190%

Correct Answer: Select (b)
Explanation: The relationship between noise dose and TWA is: $$TWA = 16.61 \log \frac{D}{100} + 90$$ where:

- TWA = time weighted noise exposure in dB
- D = noise dose in percent. For the problem at hand: $$95 = 16.61 \log \frac{D}{100} + 90$$ $$\log \frac{D}{100} = \frac{5}{16.61} = 0.301$$ $$\frac{D}{100} = 1.99 \implies D = 199%$$

Question 61

Question: The average velocity pressure in the duct of a hood is 0.70 inches of water gauge. If the hood static pressure measures 1.30 inches of water gauge, the hood entry coefficient is equal to:

- a. 9.1
- b. 0.91
- c. 0.80
- d. 0.73

Correct Answer: Select (d)
Explanation: The relationship between hood entry coefficient, velocity pressure, and static pressure is: $$C_e = \sqrt{\frac{VP}{SP_h}}$$ where:

- C_e = hood entry coefficient (dimensionless)
- VP = velocity pressure in inches of water
- SP_h = absolute value of static pressure in inches of water For the problem at hand we have: $$C_e = \sqrt{\frac{0.70}{1.30}} = 0.73$$

Question 62

Question: The velocity pressure for a hood which has a coefficient of entry equal to 0.85 and a static pressure equal to 1.1 inches of water is:

- a. 0.25
- b. 0.79
- c. 0.40
- d. 0.65

Correct Answer: Select (b)
Explanation: $$C_e = \sqrt{\frac{VP}{SP_h}}$$ $$0.85 = \sqrt{\frac{VP}{1.1}}$$ $$0.72 = \frac{VP}{1.1} \implies VP = 0.79\text{ inches of water}$$

Question 63

Question: The velocity pressure for a hood with a coefficient of entry of 0.70 measures 0.90 inches of water. The static pressure for this hood is:

- a. 1.84 in. water
- b. 0.63 in. water
- c. 1.28 in. water
- d. 0.73 in. water

Correct Answer: Select (a)
Explanation: $$C_e = \sqrt{\frac{VP}{SP_h}}$$ $$0.70 = \sqrt{\frac{0.9}{SP_h}}$$ $$0.49 = \frac{0.9}{SP_h} \implies SP_h = \frac{0.9}{0.49} = 1.84\text{ in. water}$$

Question 64

Question: In a manufacturing facility a chemical with a TLV of 200 ppm is vaporized at the rate of 0.5 cfm. In order to keep the concentration of this chemical at or below the TLV level, the minimum dilution ventilation rate must be:

- a. 100 cfm
- b. 400 cfm
- c. 4,000 cfm
- d. 2,500 cfm.

Correct Answer: Select (d)
Explanation: We use the following relationship to solve this problem: $$Q = 10^6 \times \frac{G}{TLV}$$ where:

- Q = the dilution rate in cubic feet per minute
- TLV = the threshold limit value in parts per million
- G = the rate of generation of vapor or gas in cubic feet per minute For the problem at hand, we have: $$Q = \frac{0.5}{200} \times 10^6 = 2,500\text{ cfm}$$

Question 65

Question: A volatile chemical with a TLV of 50 ppm is used in a laboratory which has an effective dilution ventilation rate of $4,000\text{ cfm}.$ The maximum allowable vapor generation for this chemical is:

- a. 0.8 cfm
- b. 0.5 cfm
- c. 0.2 cfm
- d. 0.9 cfm

Correct Answer: Select (c)
Explanation: $$Q = 10^6 \times \frac{G}{TLV}$$ $$G = \frac{Q \times TLV}{10^6} = \frac{4000 \times 50}{10^6} = 0.2\text{ cfm}$$

Question 66

Question: In a manufacturing facility the ventilation system is designed to generate an effective dilution ventilation rate of $3,000\text{ cfm}.$ The rate of vapor generation of a chemical is expected to be 0.75 cfm. The TLV of this chemical must not exceed:

- a. 50 ppm
- b. 250 ppm
- c. 1,000 ppm
- d. 750 ppm.

Correct Answer: Select (b)
Explanation: $$Q = 10^6 \times \frac{G}{TLV}$$ $$TLV = 10^6 \times \frac{G}{Q} = 10^6 \times \frac{0.75}{3000} = 250\text{ ppm}$$

Question 67

Question: The effective ventilation rate for an actual ventilation rate of $16,000\text{ cfm}$ with a factor of safety of 8 is:

- a. 128,000 cfm
- b. 48,000 cfm
- c. 2,000 cfm
- d. 4,000 cfm

Correct Answer: Select (c)
Explanation: We use the following relationship: $$Q' = \frac{Q}{C}$$ where:

- Q' = effective ventilation rate
- Q = actual ventilation rate

- C = factor of safety For the problem at hand, we have: $$Q' = \frac{16000}{8} = 2,000\text{ cfm}$$

Question 68

Question: A hazardous chemical is stored in a room at sea level where barometric pressure is $760\text{ mm Hg}.$ The vapor pressure of this chemical at 75°F is equal to $200\text{ mm Hg}.$ The saturation concentration of this chemical in the room at 75°F is:

- a. 15,000 ppm
- b. 57,000 ppm
- c. 162,943 ppm
- d. 263,158 ppm

Correct Answer: Select (d)
Explanation: We use the following relationship: $$C = \frac{10^6 \times P_v}{P_b}$$ where:

- C = saturation concentration (ppm)
- P_v = vapor pressure at a given temperature (mm Hg)
- P_b = barometric pressure (mm Hg) For the problem at hand, we have: $$C = \frac{10^6 \times 200}{760} = 263,158\text{ ppm}$$

Question 69

Question: It has been determined that the saturation concentration of a chemical at 80°F is $123,000\text{ ppm}.$ If the barometric pressure is $760\text{ mm Hg},$ the vapor pressure of this chemical at 80°F is:

- a. 62.51 mm Hg
- b. 93.48 mm Hg
- c. 12.30 mm Hg
- d. 76.25 mm Hg

Correct Answer: Select (b)
Explanation: $$C = \frac{P_v \times 10^6}{P_b}$$ $$123,000 = \frac{P_v \times 10^6}{760} \implies P_v = 93.48\text{ mm Hg}$$

Question 70

Question: In a ventilation system, the static pressure on the outlet side of the fan is equal to $0.7\text{ in. wg}.$ If the static pressure on the inlet side of the fan is 3.5 in. wg below atmosphere and the velocity pressure is $1.5\text{ in. wg},$ the fan static pressure is equal to:

- a. 2.7 in. wg
- b. 5.0 in. wg
- c. 1.3 in. wg
- d. 4.2 in. wg

Correct Answer: Select (a)
Explanation: We use the following relationship: $$SP_{\text{fan}} = SP_{\text{out}} - SP_{\text{in}} - VP_{\text{in}}$$ where:

- SP_{fan} = fan static pressure (inches of water gauge)
- SP_{out} = static pressure on the outlet side of the fan (in. wg)
- SP_{in} = static pressure on the inlet side of the fan (in. wg)
- VP_{in} = velocity pressure on the inlet side of fan (in. wg) For the problem at hand, we have: $$SP_{\text{fan}} = 0.7 - (-3.5) - 1.5 = 2.7\text{ in. wg}$$

Question 71

Question: A hood has a square opening with each side equal to 6 in. The hood has been designed to have a volumetric flow rate of $16,000\text{ cfm}.$ This hood is to be used to remove contaminant vapors 5 feet away from the opening. The capture velocity at 5 feet must be at least:

- a. 30.23 fpm
- b. 25.12 fpm
- c. 63.93 fpm
- d. 86.42 fpm

Correct Answer: Select (c)
Explanation: We use the following relationship: $$V = \frac{Q}{10 x^2 + A}$$ where:

- V = capture velocity at distance x from the hood opening (feet per minute)
- Q = volumetric flow rate of air through the system (cfm)
- x = centerline distance from the hood opening (ft.)
- A = hood opening area (ft^2) For the problem at hand, we have: $$A = 6 \times 6 = 36\text{ in}^2 = 0.25\text{ ft}^2$$ $$V = \frac{16000}{10(5)^2 + 0.25} = \frac{16000}{250.25} = 63.93\text{ fpm}$$

Question 72

Question: A hood has a circular opening with a diameter equal to 10 inches. The hood is to be used in a laboratory to remove contaminant vapors at a distance of 4 feet away from the hood opening with a capture velocity of 200 fpm. The volumetric flow rate of air through this system must at least be:

- a. 2,000 cfm
- b. 8,000 cfm
- c. 6,123 cfm
- d. 32,108 cfm

Correct Answer: Select (d)
Explanation: $$V = \frac{Q}{10 x^2 + A} \implies Q = V(10 x^2 + A)$$ $$V = 200\text{ fpm}, \quad x = 4\text{ ft}$$ $$A = \frac{\pi d^2}{4} = \frac{\pi(10)^2}{4} = 78.5\text{ in}^2 = 0.545\text{ ft}^2$$ $$Q = 200 (10 \times 16 + 0.545) = 32,109\text{ CFM} \approx 32,108\text{ CFM}$$

Question 73

Question: The hood entry loss for a hood with a duct velocity pressure equal to 0.90 inches of water gauge (in. wg) and a hood entry coefficient equal to 0.80 is equal to:

- a. 0.5 in. wg
- b. 0.72 in. wg
- c. 0.80 in. wg
- d. 0.64 in. wg.

Correct Answer: Select (a)
Explanation: We use the following relationship: $$h_e = \frac{(1 - C_e^2) VP}{C_e^2}$$ where:

- C_e = hood entry coefficient (dimensionless)
- h_e = hood entry loss (in. wg)
- VP = velocity pressure (in. wg) For the problem at hand, we have: $$h_e = \frac{(1 - 0.80^2) 0.9}{(0.80)^2} = 0.5\text{ in. wg}$$

Question 74

Question: A hood with an entry coefficient of 0.72 and a static pressure equal to 2.5 in. wg has an average air velocity through the duct equal to:

- a. 3,622 fpm
- b. 4,559 fpm
- c. 3,856 fpm
- d. 5,268 fpm

Correct Answer: Select (b)
Explanation: We use the following relationship: $$V = 4005 C_e \sqrt{SP_h}$$ where:

- V = average velocity of air through duct (fpm)
- C_e = hood entry coefficient (dimensionless)
- SP_h = hood static pressure (in. wg) For the problem at hand, we have: $$V = 4005 \times 0.72 \sqrt{2.5} = 4,559\text{ fpm}$$

Question 75

Question: The static pressure for a hood with average air velocity of 5,000 fpm and coefficient of entry equal to 0.8 is equal to:

- a. 1.72 in. wg
- b. 4.00 in. wg
- c. 2.43 in. wg
- d. 3.85 in. wg

Correct Answer: Select (c)
Explanation: $$V = 4005 C_e \sqrt{SP_h}$$ $$\sqrt{SP_h} = \frac{V}{4005 C_e}$$ $$SP_h = \left(\frac{V}{4005 C_e}\right)^2$$ $$V = 5,000\text{ fpm}, \quad C_e = 0.80$$ $$SP_h = \left(\frac{5000}{4005 \times 0.08}\right)^2 = 2.43\text{ in. wg}$$

Question 76

Question: All of the following statements are true in regard to the instrument called a "velometer" EXCEPT:

- a. It measures air velocity
- b. It measures highly sensitive electrical voltage
- c. It can be useful in conducting energy audits
- d. It can be useful in hood and spray booth monitoring

Correct Answer: Select (b)
Explanation: Statement (b) is not true in regard to a velometer. This device can also be useful in OSHA compliance checks, in evaluation of drying systems, in evaluation of fan and blower performance, and in static pressure measurement.

Question 77

Question: You are standing 2 feet from a radiation source. How far should you move away to reduce your exposure to 1/100 of the original value?

- a. 10 feet
- b. 5 feet
- c. 20 feet
- d. 40 feet

Correct Answer: Select (c)
Explanation: Radiation intensity follows the inverse square law, which means that the intensity of radiation decreases with the square of the distance form the source. $$I_1 d_1^2 = I_2 d_2^2$$ $$\frac{I_2}{I_1} = 0.01 = \frac{d_1^2}{d_2^2}$$ $$0.01 = \frac{4}{d_2^2} \implies d_2 = 20\text{ ft.}$$

Question 78

Question: The amount of radiation produced by a source at an unknown location away from the source is 500 rems. The radioactivity measured at 10 meters from the source is 50 rems. At what distance from the source the activity of 500 rems was measured?

- a. 10.50 meters
- b. 3.16 meters
- c. 6.32 meters
- d. 8.75 meters

Correct Answer: Select (b)
Explanation: $$I_1 d_1^2 = I_2 d_2^2$$ $$\frac{I_2}{I_1} = \frac{50}{500} = \frac{d_1^2}{100}$$ $$d_1 = 3.16\text{ meters}$$

Question 79

Question: A radioisotope emitting gamma rays has a half-life of 500 years. How long would it take to reduce the activity of this material to less than 15 percent of its original value?

- a. 1,000 years
- b. 1,500 years
- c. 2,000 years
- d. 2,500 years

Correct Answer: Select (b)
Explanation: | Time (Years) | Activity (% of Original) | | :--- | :--- | | 0 | 100.0 | | 500 | 50.0 | | 1,000 | 25.0 | | 1,500 | 12.5 |

At 1,500 years, activity is 12.5%, which is less than 15%.

Question 80

Question: A centrifugal fan used in a ventilation system delivers a flow rate of 100 cfm. What is the flow rate if the speed of the fan is tripled?

- a. 10,000 cfm
- b. 10 cfm
- c. 300 cfm
- d. 600 cfm

Correct Answer: Select (c)
Explanation: The fan laws state that flow rate is directly proportional to fan speed. $$\frac{N_1}{N_2} = \frac{Q_1}{Q_2}$$ $$\frac{100}{300} = \frac{100}{Q_2} \implies Q_2 = 300\text{ cfm}$$

Question 81

Question: If the impeller speed of a centrifugal speed fan is increased from 100 to 400 rpm; horse power will increase by a factor of:

- a. 24
- b. 16
- c. 64
- d. 8

Correct Answer: Select (c)
Explanation: The relationship between fan speed and horsepower is: $$\left(\frac{N_1}{N_2}\right)^3 = \frac{HP_1}{HP_2}$$ $$\left(\frac{100}{400}\right)^3 = \frac{1}{64} = \frac{HP_1}{HP_2} \implies HP_2 = 64\ HP_1$$

Question 82

Question: An atmosphere is generally considered to be oxygen deficient when the concentration of oxygen falls below:

- a. 21% by volume
- b. 16.5% by volume
- c. 19.5% volume
- d. 15.5% by volume

Correct Answer: Select (c)
Explanation: An atmosphere is considered to be oxygen deficient when the concentration of oxygen falls below 19.5 per cent by volume.

Question 83

Question: A liquid contains 40 percent (by weight) heptane with a $\text{TLV} = 1,600\text{ mg/m}^3$ and 60 percent (by weight) methylchloroform with a $\text{TLV} = 1,900\text{ mg/m}^3.$ The TLV for the mixture is:

- a. $1,800\text{ mg/m}^3$
- b. $1,685\text{ mg/m}^3$
- c. $2,010\text{ mg/m}^3$
- d. $1,767\text{ mg/m}^3$

Correct Answer: Select (d)
Explanation: $$TLV_{\text{mixture}} = \frac{1}{\frac{0.4}{1600} + \frac{0.6}{1900}} = 1767\text{ mg/m}^3$$

Question 84

Question: Air contains 700 ppm of methane $(\text{CH}_4).$ How many grams of methane is in a room that has a volume of 25 cubic meters? The molecular weight of methane is 16.

- a. 16
- b. 21
- c. 11.5
- d. 5

Correct Answer: Select (c)
Explanation: $$\text{ppm} = \frac{(\text{mg/m}^3)(24.4)}{\text{M.W.}}$$ $$700 = \frac{(\text{mg/m}^3)(24.4)}{16} \implies \text{mg/m}^3 = 459\text{ mg/m}^3$$ $$459 \times 25 = 11,475\text{ mg} = 11.475\text{ gr, rounded up to } 11.5\text{ gr}$$

Question 85

Question: What is the moisture content of air on a day when the dry bulb temperature is 80°F and the relative humidity is 70%?

- a. 0.016 lb of water per pound of dry air
- b. 0.08 lb of water per pound of dry air
- c. 0.07 lb of water per pound of dry air
- d. 0.56 lb of water per pound of dry air

Correct Answer: Select (a)
Explanation: On the psychrometric chart, first locate the point where the vertical line corresponding to 80°F dry bulb temperature and the line of 70% relative humidity intersect. From this point on the chart, draw a horizontal line to the right to read the moisture content, which is 0.016 lb of water per pound of dry air.

Question 86

Question: A liquid mixture contains 40 percent (by weight) heptane, 20% methyl chloroform, and the balance perchloroethylene. Assuming that all of this liquid evaporates, and that the atmospheric composition is similar to that of the original liquid. The TLV for the gaseous mixture generated from this liquid is equal to which of the following? The TLV for individual components are: heptane TLV = $1,600\text{ mg/m}^3;$ methyl chloroform TLV = $1,900\text{ mg/m}^3;$ perchloroethylene TLV = $335\text{ mg/m}^3.$

- a. 723 mg/m^3
- b. $1,232\text{ mg/m}^3$
- c. 645 mg/m^3
- d. 957 mg/m^3

Correct Answer: Select (c)
Explanation: $$f_1 = 0.4, \quad f_2 = 0.2, \quad f_3 = 0.4$$ $$TLV_{\text{mixture}} = \frac{1}{\frac{f_1}{(TLV)_1} + \frac{f_2}{(TLV)_2} + \frac{f_3}{(TLV)*3}}$$ $$TLV*{\text{mixture}} = \frac{1}{\frac{0.4}{1600} + \frac{0.2}{1900} + \frac{0.4}{335}} = 645\text{ mg/m}^3$$

Question 87

Question: According to Raoult's Law, in a binary system, when liquid and vapor phases have reached equilibrium:

- a. The vapor has a higher concentration of the more volatile component as compared to the liquid.
- b. The liquid has a higher concentration of the more volatile component as compared to the liquid.
- c. The concentration of the less volatile component is the same in the vapor and liquid phases.
- d. Raoult's Law deals with electricity and magnetism and does not address vapor/liquid equilibria.

Correct Answer: Select (a)
Explanation: Raoult's Law deals with vapor/liquid equilibria. It states that, in a binary system, when the liquid and vapor phases have reached equilibrium, the vapor phase has a higher concentration of the more volatile (lower boiling point) component as compared to the liquid phase. This is the principle on which separation of components by distillation is based.

Question 88

Question: Fick's Law deals with:

- a. Diffusion
- b. Heat transfer
- c. Fluid flow in pipes
- d. Friction

Correct Answer: Select (a)
Explanation: Fick's Law deals with diffusion. It expresses the rate of mass transfer per unit area by diffusion as a function of concentration difference. The constant of proportionality is called "diffusivity".

Question 89

Question: The time-weighted exposure limit for worker exposure to a chemical is 100 ppm and the threshold limit value ceiling is 500 ppm. An employee is working in a laboratory where the concentration of the chemical is at a constant level of 400 ppm. How many hours can the employee work in this laboratory without being overexposed to the chemical?

- a. 4
- b. 3
- c. 2
- d. 1

Correct Answer: Select (c)
Explanation: $$\text{8-hour TWA} = (C_1 t_1 + C_2 t_2 + \dots)/8$$ $$100 = (400 t)/8 \implies t = 2\text{ hours}$$

Question 90

Question: A flammable liquid has been spilled in a laboratory (dimensions $10\text{ m} \times 20\text{ m} \times 14\text{ m}$) and has completely evaporated into the lab space. The molecular weight of the liquid is 70 and it has a specific gravity of 0.6. The concentration of vapor in the room is 5,000 ppm. How many liters of this liquid were spilled? Assume that density of water is $1,000\text{ g/L}.$

- a. 66.8
- b. 78.7
- c. 54.6
- d. 48.8

Correct Answer: Select (a)
Explanation: $$\text{ppm} = \frac{C(\text{mg/m}^3)(24.45)}{\text{M.W.}} \implies \text{mg/m}^3 = \frac{(C \text{ in ppm})(\text{M.W.})}{24.45}$$ $$\text{mg/m}^3 = \frac{(5,000)(70)}{24.45} = 14,315\text{ mg/m}^3 = 14.315\text{ g/m}^3$$ $$\text{Volume of lab} = 10 \times 20 \times 14 = 2,800\text{ m}^3$$ $$\text{Mass} = 14.315 \times 2,800 = 40,082\text{ g}$$ $$\text{Density} = 0.6 \times 1,000\text{ g/L} = 600\text{ g/L}$$ $$\text{Volume} = \frac{40,082}{600} = 66.8\text{ L}$$

Question 91

Question: Which of the following tests is used to evaluate the mutagen-producing capability of a chemical?

- a. Carcinogen test
- b. Muta assay
- c. Ames test
- d. DNA test

Correct Answer: Select (c)
Explanation: The Ames test is used to evaluate a chemical's mutagen-producing capability.

Question 92

Question: A relatively new instrument has been given the name "Cutie Pie" and the user acceptance has been more enthusiastic than any previous instrument of this type. Which of the following is correct regarding Cutie Pie?

- a. It is a portable beta and gamma radiation meter.
- b. It measures the transmission or absorption of liquids or solids as a function of wavelength.
- c. It is an instrument used for measuring air flow and velocity for hood design.
- d. It is an instrument that is capable of analyzing a mixture and separating the mixture's components according to their molecular weights.

Correct Answer: Select (a)
Explanation: Cutie Pie is a portable beta and gamma radiation meter of exceedingly small dimensions and weight. This instrument has been named Cutie Pie due to its diminutive size.

Question 93

Question: All of the following are forms of penetrating radiation EXCEPT:

- a. X-rays
- b. Gamma rays
- c. Alpha rays
- d. Beta rays

Correct Answer: Select (c)
Explanation: Alpha rays have a very weak penetrating power. Even a sheet of paper can stop their flow. They are most dangerous when inhaled or ingested.

Question 94

Question: Which of the following instruments is used to measure noise that is variable over a time period?

- a. Sound level meter
- b. Spectrum analyzer
- c. Frequency analyzer
- d. Integrating meter

Correct Answer: Select (d)
Explanation: Integrating meters are used to measure noise that is variable over a period of time. Spectrum analyzers measure the frequency of a sound wave.

Question 95

Question: Ten gallons of a chemical with a molecular weight of 60 and a TLVC (threshold limit value ceiling) of 500 ppm is released into a $30\text{ ft} \times 30\text{ ft} \times 15\text{ ft}$ laboratory. Specific gravity is 0.6. Which of the following statements is true?

- a. TLVC has been exceeded
- b. TLVC for the 8-hour shift might not have been exceeded
- c. TLVC has not been exceeded
- d. TLVC has not been exceeded but PEL (permissible exposure limit) might have been exceeded

Correct Answer: Select (a)
Explanation: $$V = \text{room volume} = 30 \times 30 \times 15 = 13,500\text{ ft}^3 = 382.3\text{ m}^3$$ Convert

gallons to mg: $$10\text{ gal} \times 35.3145\text{ ft}^3 / 264.17\text{ gal} \times 0.6 \times 62.4\text{ lb/ft}^3 \times 1000\text{ g}/2.20462\text{ lb} \times 1000\text{ mg}/1\text{ g} = 2.2702394 \times 10^7\text{ mg}$$ $$\text{ppm} = \frac{\text{mg/m}^3 \times 24.45}{\text{MW}} = \frac{(2.2702394 \times 10^7 / 382.3) \times 24.45}{60} = 24,198\text{ ppm}$$ Since 24,198 ppm is far greater than 500 ppm, TLVC has been exceeded.

Question 96

Question: Which of the following personal protective equipment must be used in an atmosphere that is immediately dangerous to life or health (IDLH)?

- a. Negative pressure SCBA
- b. Positive pressure SCBA
- c. SCBA with indirect ventilation ports
- d. SCBA with direct ventilation ports

Correct Answer: Select (b)
Explanation: Positive pressure SCBA eliminates the possibility of any leakage from the contaminated atmosphere into the face mask.

Question 97

Question: According to the Center for Disease Control (CDC), which of the following statements is true in regard to the chemical benzene $(\text{C}_6\text{H}_6)?$

- a. Benzene works by causing cells to not work correctly
- b. It can damage the immune system
- c. It changes levels of antibodies in the blood
- d. All of the above

Correct Answer: Select (d)
Explanation: Exposure to benzene can be detected by urine tests. Signs and symptoms of over-exposure to benzene include drowsiness, dizziness, rapid or irregular heartbeat, headaches, and, at very high levels, death.

Question 98

Question: One of the major health effects of polychlorinated biphenyls (PCBs), which were widely used in electrical transformers, is cancer. Which of the following is NOT an adverse non-cancer health effect of PCBs?

- a. Effects on the immune system
- b. Effects on the reproductive system
- c. Effects on the nervous system
- d. Effects on the digestive system

Correct Answer: Select (d)
Explanation: Even today, PCBs can be found in old electrical transformers. According to the EPA, statements (a), (b), and (c) are non-cancer adverse health effects of PCBs.

Question 99

Question: Which of the following statements is correct in regard to bloodborne pathogens described in the OSHA bloodborne pathogen standard (29 CFR 1910.1030)?

- a. They are pathogenic microorganisms that are present in human blood and can cause disease in humans
- b. Hepatitis B virus (HBV) is an example of these pathogens
- c. Human immunodeficiency virus (HIV) is an example of these pathogens
- d. All of the above

Correct Answer: Select (d)
Explanation: Statements (a), (b), and (c) are true in regard to bloodborne pathogens.

Question 100

Question: According to the Center for Disease Control (CDC), there are four biosafety levels (BSL-1 through -4). Which of the following statements is NOT true in regard to biosafety levels?

- a. In BSL-1, the microbes are known to consistently cause disease in healthy adults.
- b. In BSL-2, the microbes pose moderate hazards to humans and the environment.
- c. In BSL-3, the microbes can cause serious or potentially lethal disease.
- d. In BSL-4, the microbes pose a high risk of infections that are frequently fatal.

Correct Answer: Select (a)
Explanation: In BSL-1, the microbes pose minimal potential hazard to humans and the environment. The personal protective equipment for a BSL-1 laboratory are lab coats, gloves, and eye protection.

Question 101

Question: Sick Building Syndrome (SBS) is a situation in which occupants of a building experience acute health effects that seem to be linked to time spent in a building, but no specific cause can be identified. Which of the following is NOT part of what causes SBS?

- a. Contaminants from outdoor sources
- b. Contaminants from indoor sources
- c. Too many glass windows
- d. Biological contaminants

Correct Answer: Select (c)
Explanation: Another cause of SBS is inadequate ventilation.

Question 102

Question: Which of the following statements is true in regard to dosimeters (film badges) for radiation monitoring?

- a. They are not capable of measuring alpha or soft beta emitters with an average energy less than 100 keV.
- b. Only the person who is assigned a film badge shall wear it.
- c. Dosimeters should be worn whenever there is a possibility of being exposed to ionizing radiation.
- d. All of the above.

Correct Answer: Select (d)
Explanation: Statements (a), (b), and (c) are all true in regard to dosimeters.

Question 103

Question: Q fever is a worldwide disease with acute and chronic stages caused by the bacteria Coxiella burnetii. Which of the following animals are the primary reservoir(s) for this bacteria?

- a. Cattle
- b. Sheep
- c. Goats
- d. All of the above

Correct Answer: Select (d)
Explanation: Organisms are excreted in milk, urine, and feces of infected animals. Infection of humans usually occurs by inhalation of these organisms from air that contains contaminated airborne barnyard dust. Humans are often very susceptible to the disease, and very few organisms may be required to cause infection. The symptoms include high fevers (up to $104-105^\circ\text{F}$), severe headache, general malaise, myalgia, chills and/or sweats, non-productive cough, nausea, vomiting, diarrhea, abdominal pain, and chest pain.

Question 104

Question: Which of the following statements is NOT true in regard to the acute disease anthrax?

- a. It affects both humans and animals
- b. Most forms of the disease are lethal
- c. Some forms of the disease respond well to antibiotics
- d. There are no effective vaccines against anthrax

Correct Answer: Select (d)
Explanation: There are effective vaccines against anthrax. French scientist Louis Pasteur developed the first effective anthrax vaccine in 1881.

Question 105

Question: Which of the following statements is NOT true in regard to heat syncope?

- a. It is fainting as a result of overheating
- b. It is caused by mild overheating with inadequate water or salt
- c. It occurs when blood pressure is raised as a result of overexposure to heat
- d. It can be treated by positioning the patient in a seated position with legs raised, administering water slowly, and moving the patient to a cooler area

Correct Answer: Select (c)
Explanation: Heat syncope occurs when blood pressure is lowered as a result of overexposure to heat. Also, water is evaporated from the blood, reducing blood volume and causing an eventual further drop in blood pressure. The result is less blood to the brain, causing fainting.

Question 106

Question: Which of the following statements is NOT true in regard to infrared radiation?

- a. It is absorbed by the skin
- b. It is absorbed as heat
- c. It is absorbed by the eyes
- d. It is a form of ionizing radiation

Correct Answer: Select (d)
Explanation: Workers normally notice excessive exposure through heat sensation and pain. Sources of IR radiation include furnaces, heat lamps, and IR lasers. IR is categorized as non-ionizing radiation. The most damage it causes is to the eyes. Other forms of non-ionizing radiation include the spectrum of ultraviolet (UV), visible light, microwave (MW), and radio frequency (RF).

Question 107

Question: Lasers typically emit which of the following optical radiations?

- a. Ultraviolet (UV)
- b. Visible light
- c. Infrared radiation (IR)
- d. All of the above

Correct Answer: Select (d)
Explanation: These non-ionizing radiations are primarily an eye and skin hazard.

Question 108

Question: Which of the following statements is true in regard to Draeger tube systems?

- a. They are glass vials filled with a chemical that reacts to a specific chemical or family of chemicals
- b. A known volume of air is drawn through a tube with a pump
- c. If the targeted chemicals are present, the chemical in the tube changes color
- d. All of the above

Correct Answer: Select (d)
Explanation: Statements (a), (b), and (c) are all true in regard to Draeger tubes. For example, the reagent for very low concentrations of sulfur dioxide (0.1-3.0 ppm) changes color to orange.

Question 109

Question: Which of the following statements is true in regard to the Ames test?

- a. It is an indirect assay for potential carcinogens
- b. It uses several strains of the bacterium Salmonella typhimurium
- c. It identifies mutagens that are also possible carcinogens
- d. All of the above

Correct Answer: Select (d)
Explanation: The Ames test has been used to identify potential carcinogens used in commercial products. An example of one such substance is trisphosphate, which was used as a flame retardant in plastic and textiles used in children's sleepwear.

Question 110

Question: Which of the following diseases is caused by inhalation of metal fumes or dust generated in most welding and grinding operations?

- a. Siderosis
- b. Byssinosis
- c. Brucellosis
- d. Presbycusis

Correct Answer: Select (a)
Explanation: Byssinosis is an occupational disease that is caused by overexposure to cotton dust. Brucellosis, also called Malta fever, is caused by ingesting unsterilized milk or meat. Presbycusis is loss of hearing as a result of aging.

Question 111

Question: Which of the following statements is true in regard to threshold limit values (TLVs)?

- a. TLV-time weighted average (TWA) is based on the average exposure in an eight hours per day, 40 hours per week schedule.
- b. TLV-short term exposure limit (STEL) is spot exposure for 15 minutes and cannot be repeated more than four times a day.
- c. TLV-ceiling (TLVC) is the exposure limit that cannot be exceeded even momentarily.
- d. All of the above

Correct Answer: Select (d)
Explanation: Statements (a), (b), and (c) are true in regard to TLVs. TLVs, along with biological exposure indices (BEIs), are published annually by the American Conference of Governmental Industrial Hygienists (ACGIH).

Question 112

Question: All of the following are areas used in the design of HVAC systems EXCEPT:

- a. Heat transfer
- b. Diffusion
- c. Fluid mechanics
- d. Thermodynamics

Correct Answer: Select (b)
Explanation: Heating, ventilation, and air conditioning (HVAC) design uses the principles of heat transfer, fluid mechanics, and thermodynamics. The design must initially have an estimate of the required heating or cooling, which depends on the outside climate as well as the inside desired conditions.

Question 113

Question: Which of the following statements is true regarding demand control ventilation to maintain acceptable indoor air quality?

- a. Carbon monoxide sensors are used to control the rate of air flow.
- b. Carbon dioxide sensors are used to control the rate of air flow.
- c. Oxygen sensors are used to control the rate of air flow.
- d. All of the above

Correct Answer: Select (b)
Explanation: When the carbon dioxide exhaled by the occupants of a building reaches a certain concentration, the carbon dioxide sensors in the building activate the ventilation system.

Question 114

Question: When manually lifting a load, the load must be kept:

- a. As far away from the body as possible.
- b. As low as possible.
- c. As close to the body as possible.
- d. In such a way that the line through the center of gravity of the load makes an angle of 80 degrees with the hands.

Correct Answer: Select (c)
Explanation: The load must be kept as close to the body as possible. This would minimize the moment created by the load on the back muscles.

Question 115

Question: The moment generated on the back muscles of a woman as a result of lifting a load compared to the moment generated on the back muscles of a man lifting the same load is generally:

- a. Smaller
- b. Larger
- c. Same
- d. Depends on the shape of the load

Correct Answer: Select (b)
Explanation: A woman's torso length relative to body weight is generally larger than that for a man. This would generally create a larger value of moment on the back muscle of a woman for the same load.

Question 116

Question: In lifting a load:

- a. The back should be kept naturally straight
- b. The load must be lifted using legs
- c. The back should make a 90 degree angle with the load
- d. Only (a) and (b) above

Correct Answer: Select (d)
Explanation: It is important to keep the back naturally straight and use the legs in lifting a load. This would not generate excessive pressure on the back muscles.

Question 117

Question: The science that applies the knowledge related to human capabilities and limitations to the design of work environments is called:

- a. Anthropometry
- b. Epidemiology
- c. Ergonomics
- d. Anthropology

Correct Answer: Select (c)
Explanation: Ergonomics relates human capabilities and limitations to the design of work environment to achieve an optimum degree of efficiency and safety. Ergonomics is the science of fitting workplace conditions and job demands to the capabilities of the working population. Effective and successful "fits" assure high productivity, avoidance of illness and injury risks, and increased satisfaction among the workforce.

Although the scope of ergonomics is much broader, the term here refers to assessing those work-related factors that may pose a risk of musculoskeletal disorders and recommendations to alleviate them. Common examples of ergonomic risk factors are found in jobs requiring repetitive, forceful, or prolonged exertions of the hands; frequent or heavy lifting, pushing, pulling, or carrying of heavy objects; and prolonged awkward postures. Vibration and cold may add risk to these work conditions.

Jobs or working conditions presenting multiple risk factors will have a higher probability of causing a musculoskeletal problem. The level of risk depends on the intensity, frequency, and duration of the exposure to these conditions. Environmental work conditions that affect risk include the intensity, frequency, and duration of activities.

Question 118

Question: In designing a piece of equipment, the controls and displays must conform to which of the following principles?

- a. Controls must be positioned opposite to the displays.
- b. The movement of controls should follow the principle of non-congruity.
- c. If the display moves to the right, the control should move counterclockwise.
- d. Displays should generally move in the same direction as movement of controls.

Correct Answer: Select (d)
Explanation: Control and display movement must be compatible. (Also see description in the text.)

Question 119

Question: Some industrial tasks may cause tendonitis, bursitis, or carpal tunnel syndrome. Which one of the following is more likely to cause these diseases?

- a. One time lifting of a very heavy object
- b. Extensive use of visual display terminals
- c. A static load on hand and shoulder
- d. Frequent twisting in low back area

Correct Answer: Select (c)
Explanation: The presence of "static loading" has been found to cause or aggravate disorders such as tendonitis, bursitis, or carpal tunnel syndrome.

Question 120

Question: In occupational ergonomics, the abbreviation CTD stands for:

- a. Carpal tunnel disorders
- b. Caption translation for deaf
- c. Critical task design
- d. Cumulative trauma disorders

Correct Answer: Select (d)
Explanation: In occupational ergonomics, CTD is an abbreviation for Cumulative Trauma Disorders. Cumulative trauma disorders (CTDs) are injuries of the musculoskeletal and nervous systems that may be caused by repetitive tasks, forceful exertions, vibrations, mechanical compression (pressing against hard surfaces), or sustained or awkward positions.

Question 121

Question: To prevent overexertion from manual material handling tasks, a comprehensive and systematic approach is needed involving all elements of a work system. Which one of the following is more likely NOT to be included in such an analysis?

- a. Object's material and toxicity
- b. Object's weight
- c. Worker characteristics
- d. Available mechanical lifting aids

Correct Answer: Select (a)
Explanation: Although an object's toxicity is an important factor in safe handling of the object, it is not included as a factor for study of overexertion as a result of manual material handling.

Question 122

Question: Approximately what percent of musculoskeletal injuries are caused by pushing or pulling objects?

- a. 2%

- b. 6%
- c. 20%
- d. 40%

Correct Answer: Select (c)
Explanation: Inappropriate pushing or pulling has been responsible for approximately 20 percent of musculoskeletal injuries. Some injuries that are classified as slips or falls can also have contribution from pushing or pulling objects.

Question 123

Question: What is the approximate pushing force (in pounds) that a healthy young male can exert on an object, if the floor coefficient of friction is 0.3?

- a. 12
- b. 18
- c. 44
- d. 81

Correct Answer: Select (c)
Explanation: A healthy young male, on the average, can exert a pushing force of 44 lb when the coefficient of friction between shoes and the floor is 0.3. For coefficients of friction greater than 0.6, the mean push or pull strength increases to 66 lb for the same group.

Question 124

Question: The probability of tipping over an object by pushing it increases as the:

- a. Grasping point moves above the object's center of gravity
- b. Grasping point moves toward the worker's center of gravity
- c. Grasping point moves below the object's center of gravity
- d. Grasping point becomes more parallel to the object's center of gravity

Correct Answer: Select (a)
Explanation: Objects tip over as a result of the moment which is generated by the pushing force. The farther the grasping point from object's center of gravity, the higher the probability for the object to tip over.

Question 125

Question: In designing a task requiring pushing objects, it is advantageous to consider which of the following?

- a. Use the body weight to lift and then push the object.
- b. Use a mechanical aid as a leverage to lift the load.
- c. Use the body weight to push without any lifting force applied.
- d. Reduce the coefficient of friction to make it easy for the load to slide.

Correct Answer: Select (c)
Explanation: The safest way to push an object is described in statement (c).

Question 126

Question: Based on anthropometric data, the design features of a work station should accommodate which of the following range of measurements?

- a. 0 to 100 percent of population
- b. 5th to 95th percentile
- c. 10th to 90th percentile
- d. 50th percentile

Correct Answer: Select (b)
Explanation: Based on anthropometric data, (which is the study of dimensions, weights, and strengths of various human body members for the design of equipment and work stations), the design features of a work station should accommodate from 5th percentile female to 95th percentile male.

Question 127

Question: A piece of equipment is being designed which requires workers to stand and reach vertically to activate its emergency switch. Which one of the following do you recommend for the design of the height of the switch?

- a. The 5th percentile of female population
- b. The 95th percentile of male population
- c. The 50th percentile of average person
- d. The 90th percentile of male population

Correct Answer: Select (a)
Explanation: See description in the text.

Question 128

Question: A worker lifts finished products from an assembly line that is 20 inches above the floor and places them in boxes that are located 35 inches above the floor. The weight of each product is 15 lbs. For each lift, the distance of hands away from midpoint between ankles is 12 inches. In order to properly place the load at its destination, the worker needs to turn 45 degrees. The worker places 5 finished products in boxes every minute. The coupling type is classified as "Good." The LI at the origin and destination for 3 hours of continuous work are respectively:

- a. 1.150, 1.355
- b. 0.934, 1.722
- c. 1.349, 0.836
- d. 2.739, 0.728

Correct Answer: Select (a)
Explanation: For a detailed discussion of NIOSH revised lifting equations, see the section, "Ergonomics and Human Factors Engineering."

The lift must be analyzed at both the origin and destination of the lift:

Mult	Origin	Destination
HM	$10/H = 10/12 = 0.833$	$10/H = 10/12 = 0.833$

VM	$1 - (0.0075	20 - 30
DM	$0.82 + (1.8/15) = 0.940$	$0.82 + (1.8/15) = 0.940$
AM	$1 - (0.0032 \times 0) = 1.000$	$1 - (0.0032 \times 45) = 0.856$
FM	0.35 (from Table)	0.35 (from Table)
CM	1.00 (from Table)	1.00 (from Table)

$$\text{RWL (origin)} = 51 \times 0.833 \times 0.925 \times 0.940 \times 1.000 \times 0.35 \times 1.00 = 13\text{ lbs.}$$ $$\text{LI} = 15/13 = 1.15\text{ RWL}$$ $$\text{RWL (destination)} = 51 \times 0.833 \times 0.925 \times 0.940 \times 0.856 \times 0.35 \times 1.00 = 11.06\text{ lbs.}$$ $$\text{LI} = 15/11.06 = 1.355$$

Discussion: Since LI exceeds unity (1.0) at both the origin and the destination of the lift, this lifting is considered to be hazardous. The lift at the destination, with a Lifting Index of 1.4, is more hazardous than the lift at its origin. The lift's destination is therefore the part on the lift that deserves greatest attention.

NIOSH recommends that ergonomic redesign be used to engineer the hazards out of the lifting situation. However, in this case, the lifting could be improved (in order of importance) by these administrative controls:

- Reducing the frequency of lifting.
- Reducing the work duration from 3 hours to less than two hours of continuous lifting by shift rotation. These changes can bring the LI down to 0.57. Ergonomic redesign is also important, and much more desirable than depending upon administrative controls. By raising the line to 30 inches, and positioning the load closer, the LI is reduced to less than half of unity (0.49), significantly increasing the safety of the workers at this workstation. Then, if the workers revert to the more hazardous lifting technique of twisting to place the load, the LI still remains at just above half of unity (0.57) and their safety has not been compromised.

Question 129

Question: The term "heat stress" deals with which of the following?

- a. Too much heat gained by the body
- b. Too much heat loss from the body
- c. Both heat loss from or heat gained by the body
- d. When body temperature is above 110°F

Correct Answer: Select (a)
Explanation: Heat stress refers to too much heat gained by the body. In general, the human body is more capable of handling the heat loss from the body compared to heat gain. Heat stress is determined by calculating the Wet Bulb Globe Temperature (WBGT) and comparing it to the threshold values published by the ACGIH.

Question 130

Question: Illumination is measured in which of the following units?

- a. Foot-candle
- b. Light index
- c. Lux
- d. Illuminance

Correct Answer: Select (c)
Explanation: The "Lux" is the SI unit of illuminance measuring luminous flux per unit area. It is equal to one lumen per square meter.

Question 131

Question: Which of the following statements is the correct definition of "dose"?

- a. Time-weighted TLV
- b. Short term exposure limit
- c. Concentration multiplied by exposure duration
- d. Slope of the dose-response curve

Correct Answer: Select (c)
Explanation: Statement (c) is the correct definition of dose.

Question 132

Question: Which of the following best describes the science of epidemiology?

- a. It deals with prevention of disease in a population.
- b. It deals with frequency of diseases in a population.
- c. It deals with distribution of diseases in a population.
- d. All of the above.

Correct Answer: Select (d)
Explanation: Descriptive epidemiological studies focus on the number of cases of a disease in a given time period. This type of study provides information on the distribution of the disease as well as the risk factors for an individual to contract the disease.

Question 133

Question: Which of the following is the correct description of "Presbycusis"?

- a. A condition caused as a result of overexposure to metal fumes.
- b. An irreversible hearing loss caused by damage to organ of "corti" and degeneration of neural elements of the auditory nerve.
- c. Loss of sensitivity to high frequency noise as a result of aging
- d. Lack of ability of a person to interpret what is heard.

Correct Answer: Select (c)
Explanation: Statement (b) is a description of "Sensorineural Impairment" and statement (d) is a description of "Central Impairment".

Question 134

Question: The type of radiation which is capable of dislodging electrons from the atoms is called:

- a. Laser radiation.
- b. Ultraviolet radiation.

- c. Ionizing radiation.
- d. All of the above.

Correct Answer: Select (c)
Explanation: Ionizing radiation causes dislodging of electrons. When electrons are removed from an atom which is neutral in its normal state, ions are formed, and therefore, the term ionizing radiation.

Question 135

Question: A radioactive material has a half-life of 200 days. If we start with 500 grams of this material, how many grams is left after 450 days?

- a. 250 grams
- b. 210 grams
- c. 325 grams
- d. 105 grams

Correct Answer: Select (d)
Explanation: $$N(t) = N_0(1/2)^{t/t_{1/2}}$$ Where:

- N_0 is the initial quantity of the substance that will decay (500 grams),
- $N(t)$ is the quantity that still remains and has not yet decayed after a time t (450 days),
- $t_{1/2}$ is the half-life of the decaying quantity (200 days). $$N(t) = 500 \times (1/2)^{450/200} = 500 \times (1/2)^{2.25} = 105\text{ grams}$$

Question 136

Question: Which of the following statements is correct regarding nanoparticles?

- a. Are particles between 1 and 100 nanometers (nm) in size with a surrounding interfacial layer.
- b. Through nanotechnology, nanoparticles can help make materials lighter, more durable, and more reactive.
- c. Naturally occurring nanoparticles can be found in volcanic ash, ocean spray, fine sand and dust, and even biological matter (e.g. viruses).
- d. All of the above.

Correct Answer: Select (d)
Explanation: Nanoparticles are particles between 1 and 100 nanometers (nm) in size with a surrounding interfacial layer. In nanotechnology, a particle is defined as a small object that behaves as a whole unit with respect to its transport and properties. Particles are further classified according to diameter.

Through nanotechnology, nanoparticles can help make materials lighter, more durable, and more reactive. Nanoparticles are applicable in a wide variety of industries and uses, including electronics to make computers faster, memory chips smaller, and screens clearer and brighter.

Naturally occurring nanoparticles can be found in volcanic ash, ocean spray, fine sand and dust, and even biological matter (e.g. viruses).

Question 137

Question: Employees who abuse alcohol and drugs can pose significant safety and other problems for their employers, managers, and co-workers. Some of the signs of drug and alcohol abuse are diminished job performance, lower productivity, absenteeism, tardiness, mood changes, and problems with relationships. You suspect that one of your employees is abusing drugs. Your company's drug and alcohol testing policy does not include drug testing for reasonable suspicion. All the following are among the actions you should take EXCEPT:

- a. Document any complaints, concerns, behavior patterns or witnesses to the behavior.
- b. Observe employee's behaviors with a second management person.
- c. Meet with the employee.
- d. Ask the employee to agree to a drug test to avoid disciplinary action.

Correct Answer: Select (d)
Explanation: Many times, the coworkers and /or clients notice an employee being under influence before a supervisor notices the same. It is extremely important to document any complaints, concerns, behavior patterns or witnesses to the behavior. It is important that a second management person also observes the behavior. Both observers must document their observations in detail. Only after everything is documented, you should meet with the employee and discuss what has been observed. Always have a second party at the meeting as a witness. Once these steps have been taken, you have the option of enforcing whatever is stated in the company's drug and alcohol policy.

Question 138

Question: According to the results of a National Institute for Drug Abuse (NIDA) sponsored survey, which of the following statements is true?

- a. Drug-using employees are 3.6 times more likely to be involved in a workplace accident.
- b. Drug-using employees are 2.5 times more likely to have absences of eight days or more.
- c. Drug-using employees are three times more likely to be late for work.
- d. All the above.

Correct Answer: Select (d)
Explanation: Also, drug-using employees are 2.2 times more likely to request early dismissal or time off, and five times more likely to file a workers' compensation claim.

Question 139

Question: Which of the following are signs that an employee might be abusing drugs?

- a. Frequent tardiness and several unexplained absences.
- b. Inconsistency on job performance.
- c. Causing several small incidents which resulted in minor injuries.
- d. All the above.

Correct Answer: Select (d)
Explanation: Other signs of drug abuse include showing very little concern over personal appearance and practicing very poor hygiene, unusual physical symptoms or behaviors such as hyperactivity, paranoia or overreacting to constructive criticism, and lower levels of productivity in the mornings.

Question 140

Question: Any organization that enters a federal contract for the procurement of property or services valued at $100,000 or more or receives any federal grant must comply with the regulations of which of the following acts?

- a. Consumer Products Safety Act
- b. Federal Water Pollution Control Act
- c. Drug-Free Workplace Act
- d. Clean Air Act

Correct Answer: Select (c)
Explanation: The 1988 Drug-Free Workplace Act requires certification to the federal agency issuing the contract that the organization will provide a drug-free workplace. It must be pointed out that although legal in several states, marijuana remains a schedule I drug under federal Controlled Substance Act, and therefore, employers must include marijuana prohibition in their workplace drug control policy. It must also be pointed out that many jurisdictions allow employee drug testing based on probable cause.